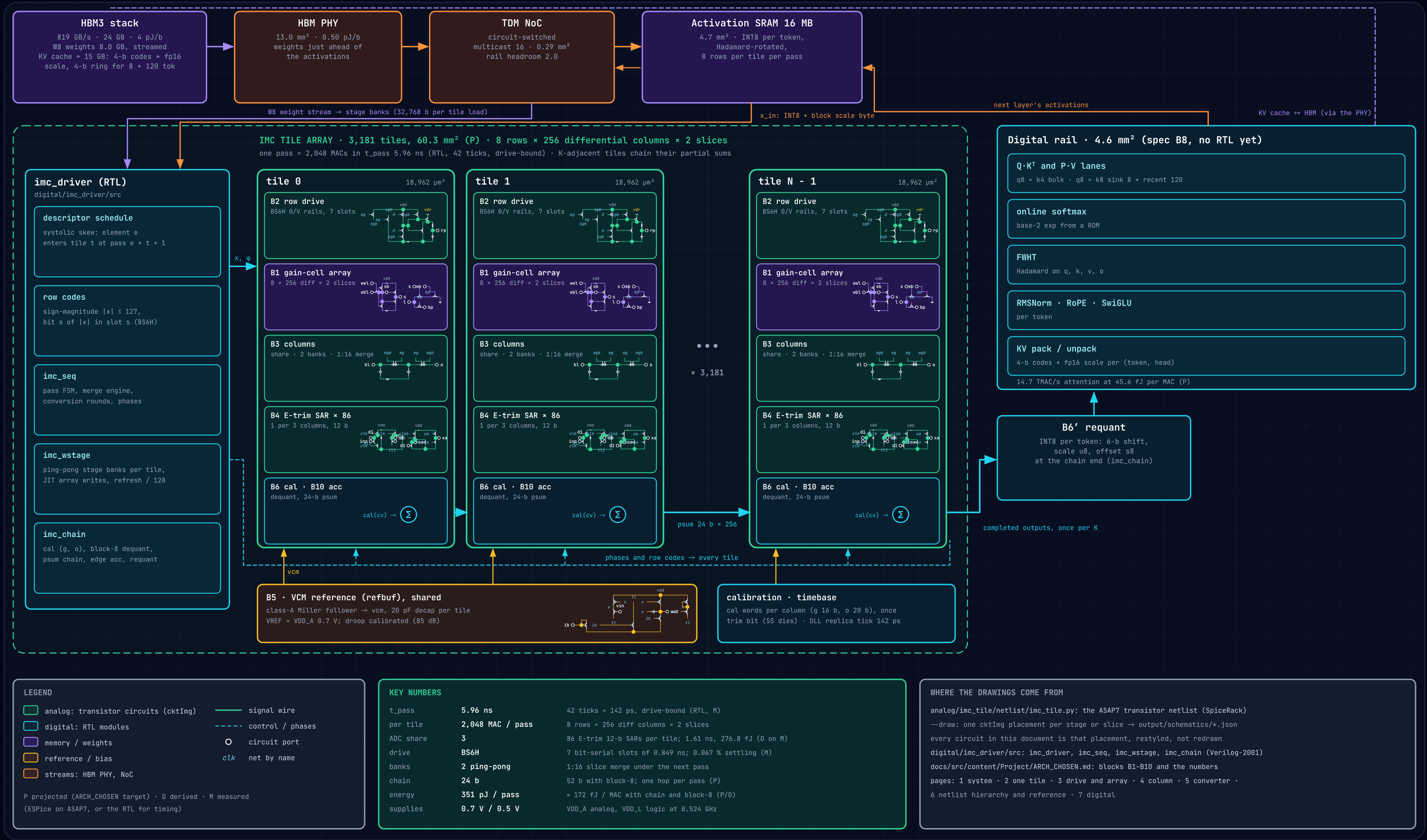
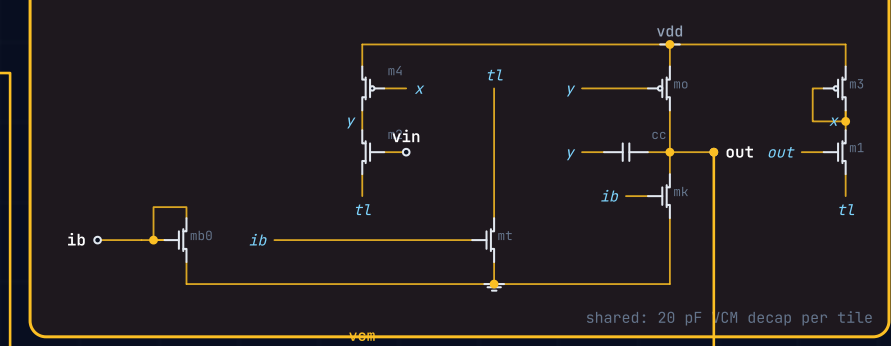
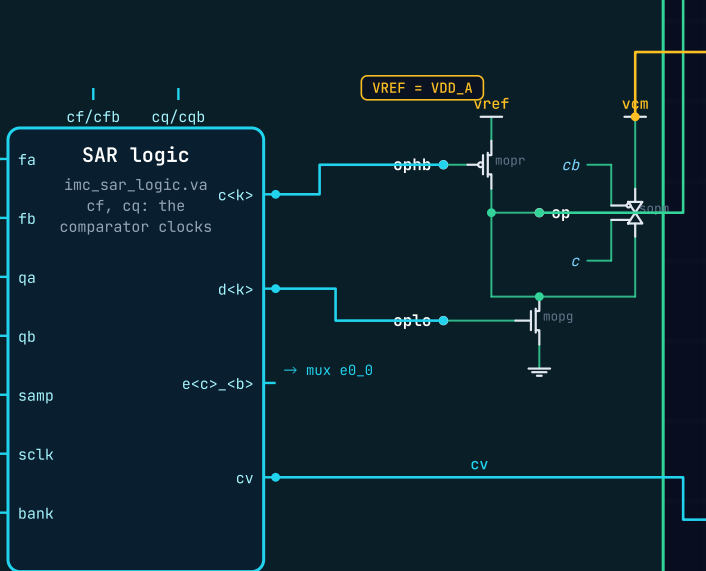
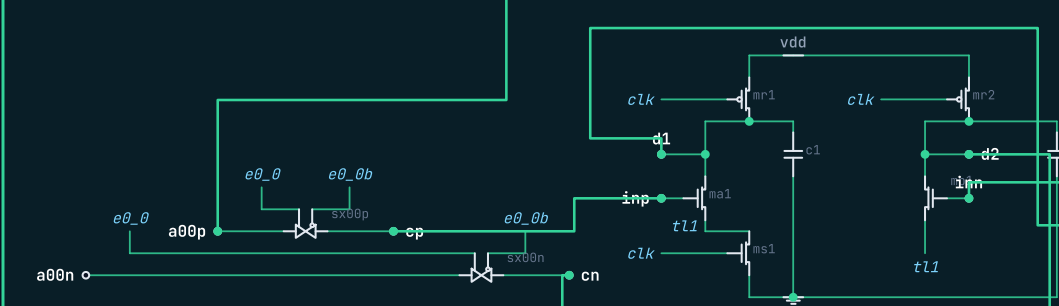
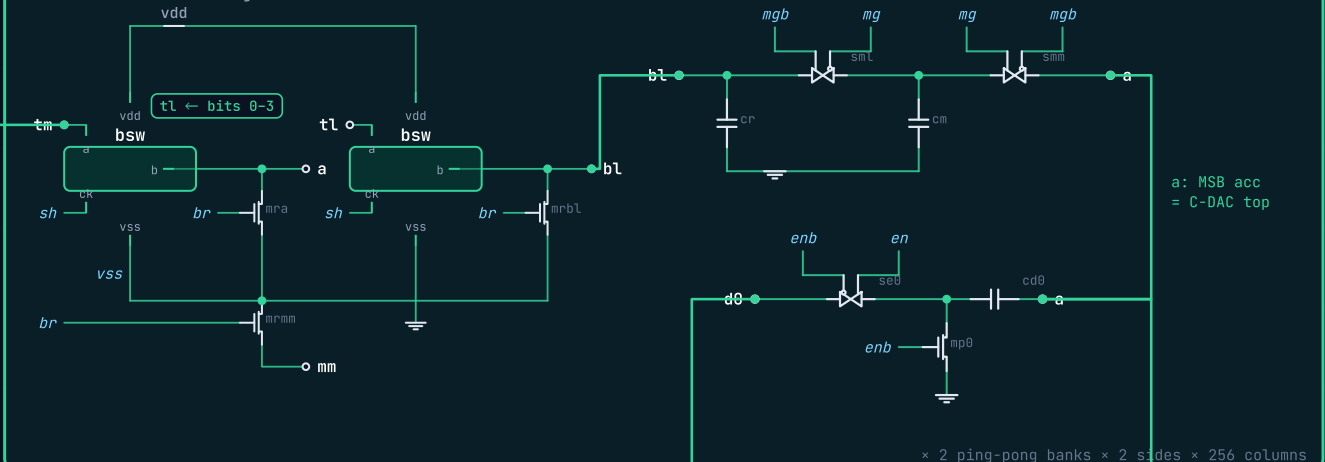
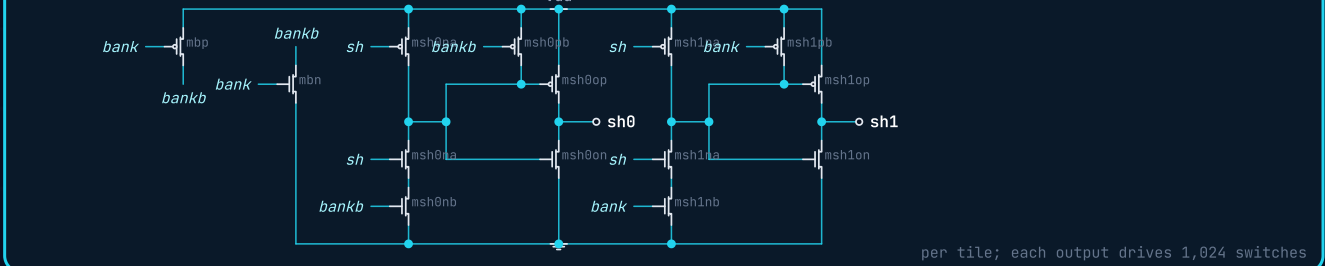
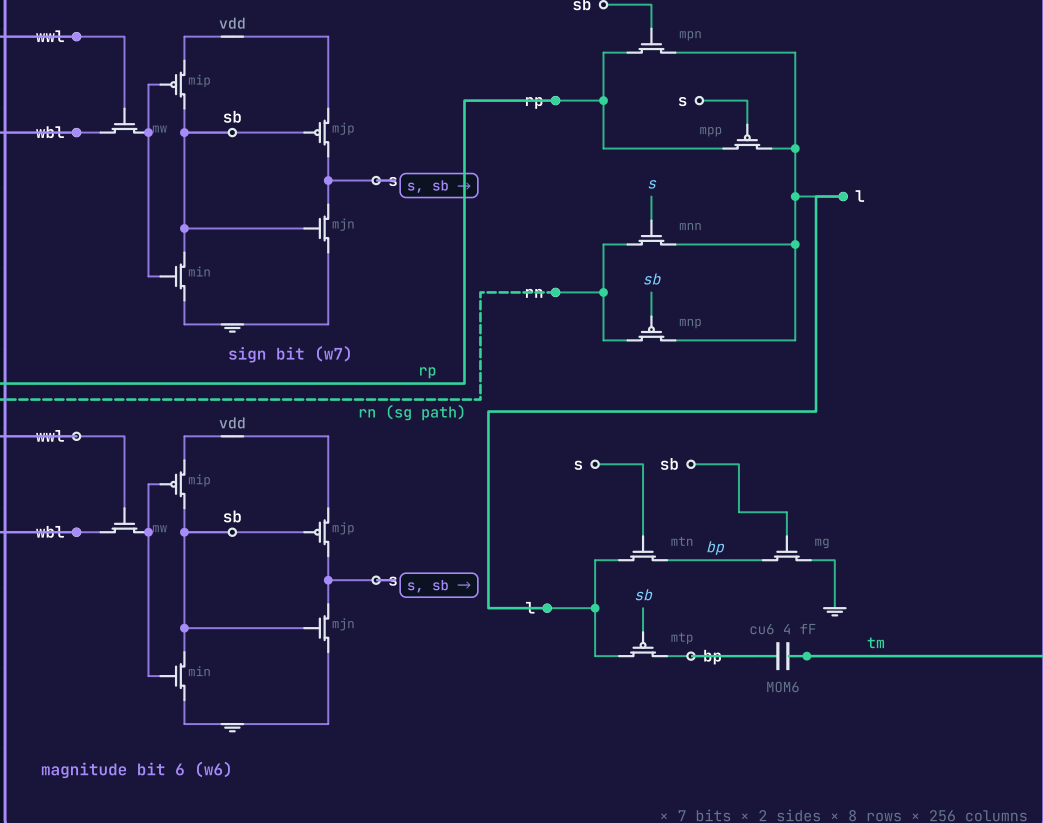
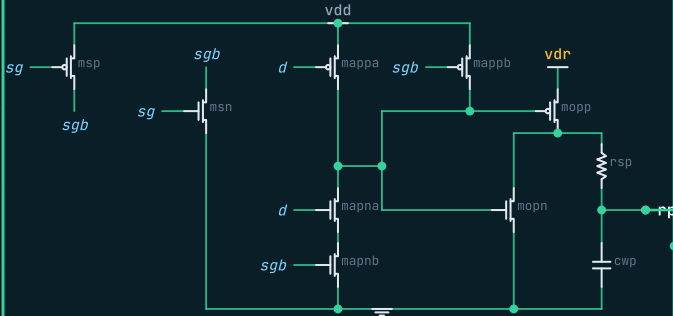


HBM → NoC → IMC tile array (K-adjacent accumulator chain) → requant → digital rail. Every tile part carries its transistor circuit, placed by cktImg from imc_tile.py.



imc_driver · row codes
drive_o: {sgn, bit s of |x|} per row
d sg



```

cv: 12-b code per column, 3 rounds per pass
B6 cal: (g.code + o + 2^13) >> 14
block-8 dequant: (c.ox.ow) << (14 - ex - ew)
B10: psum += cal (24 b; 52 b block-8)
→ the K-adjacent tile; requant at the end

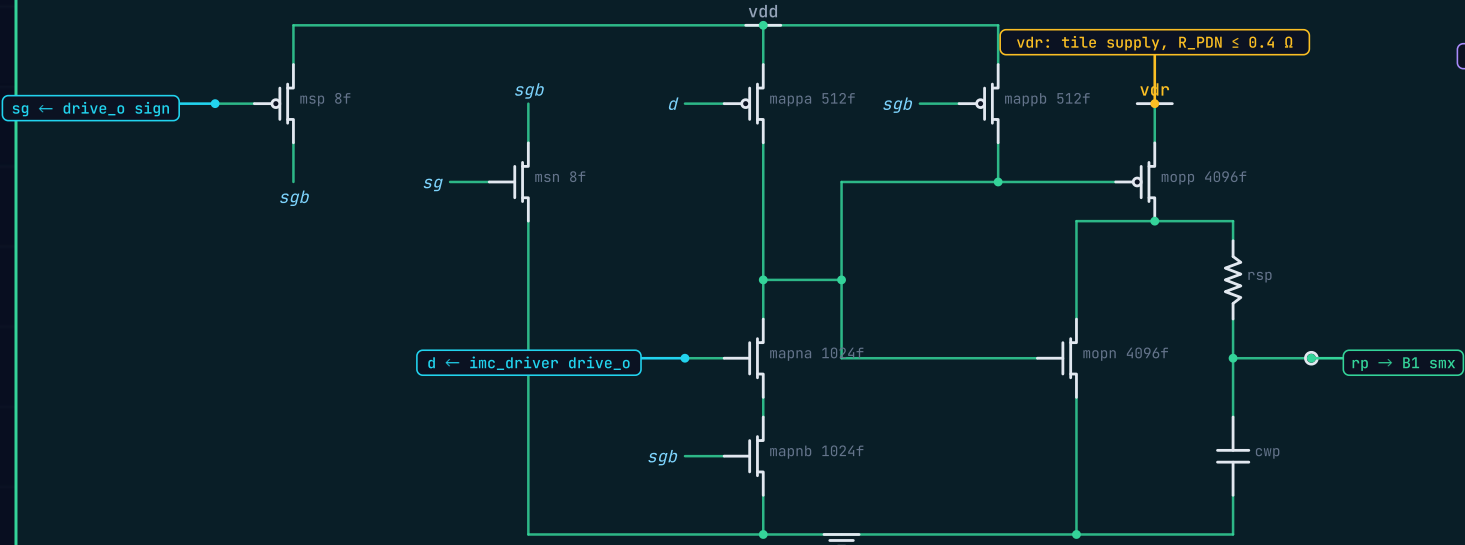
```

● B2 row drive and B1 gain-cell array

The bit-serial driver, the gain bit, the sign mux and the crosspoint, and the block views that compose them into a weight, a row and the array.

B2 · bit-serial row driver (rdrv)

BS6H: slot s puts bit s of |x| on rp (x ≥ 0) or rn (x < 0) as 0 or V

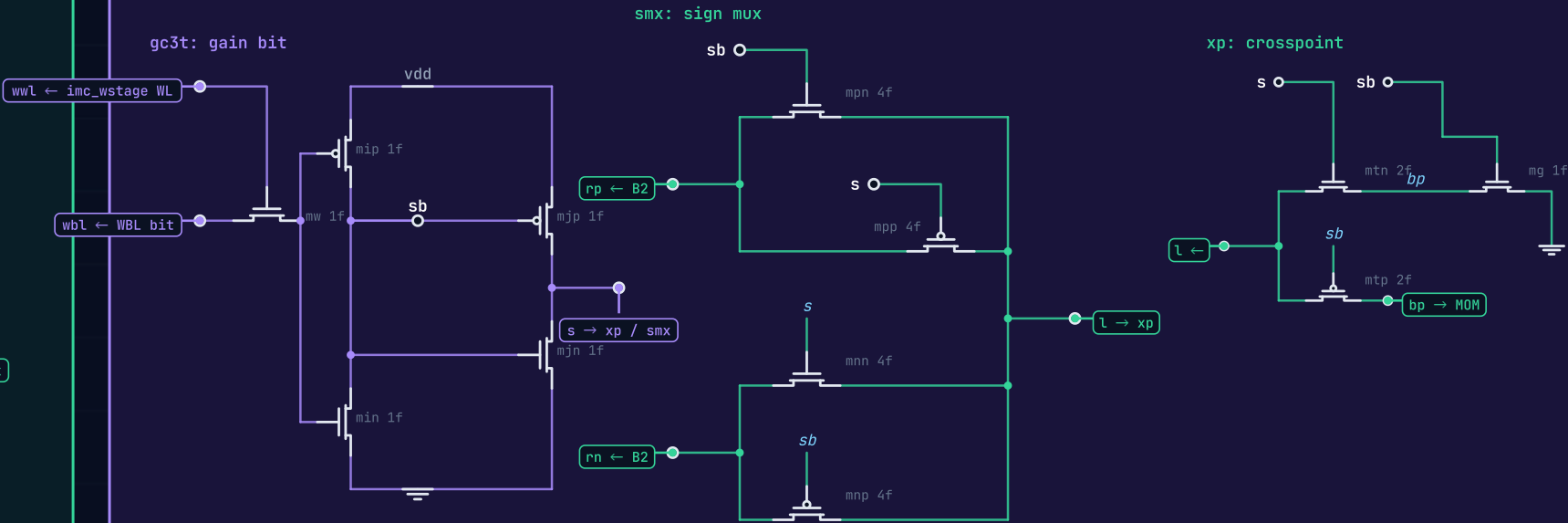


The NAND passes the bit when the sign is right: rp carries d when sg = 0, rn when sg = 1.
The rail inverter is 4,096 fins for a full 256-column row (7.47 pF of plates).
rsp is the segment strap (46 Ω / 128), cwp the rail wire (0.3 pF).
Settling at the share edge, 8 rows on V: 0.067 % (M, ESpice, with the share kick).
Fallback BS6H-cc adds a constant-charge dummy (not drawn): 3 % tracking residual.

netlist: rdrv · drawn: the rp rail; × 8 rows per tile

B1 · gain bit, sign mux, crosspoint

one W8 weight side = 8 gain bits (sign + 7) + 1 sign mux + 7 crosspoints + 7 MOM6 units



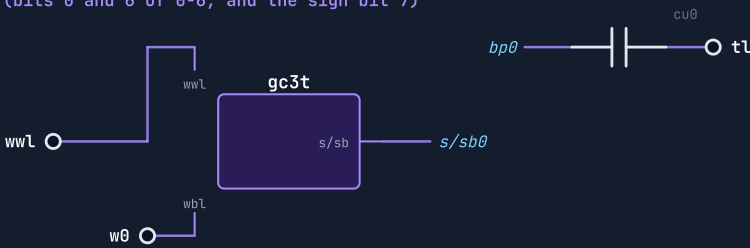
gc3t: SRAM-Vt write FET onto node m (held as charge); two inverters give s and sb (the second inverter is the gc5t buffer: 3.3 % plate error without it, M).
smx: the side's line l follows rp (w ≥ 0) or rn (w < 0); the - side swaps rp / rn.
xp: bit '1' → the T6 ties the unit's bottom plate to l; '0' → mg grounds it.
Units: MSB slice 1/2/4 × 1 fF, LSB slice 1/2/4/8 × 0.25 fF, top plates tm / tl.

netlist: gc3t, smx, xp

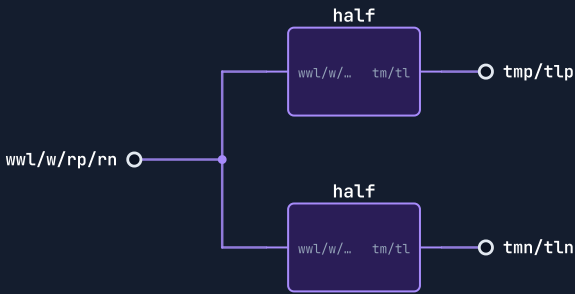
How they compose (cktimg block views of the netlist, buses bundled)

half: one side of one weight · wcell: the two sides · rows: the drivers · arr: the array

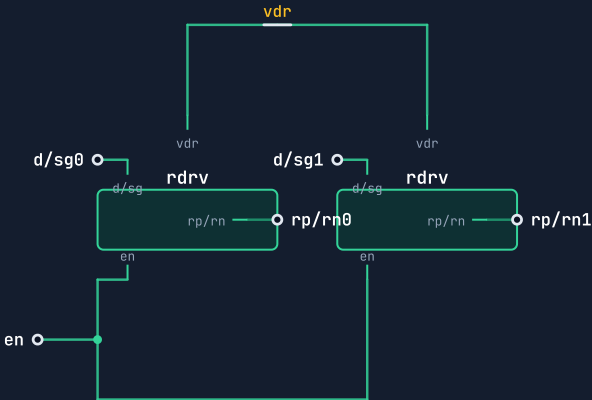
half (bits 0 and 6 of 0-6, and the sign bit 7)



wcell = 2 × half (the - side on swapped rails)



rows = 8 × rdrv (2 of 8 rows)



arr = 8 rows × 256 wcell: row r's rp/rn and wwl run along the row, column c's tmp / tmn / tlp / tln (the slice top plates) down the column.
Per tile: 4,096 half-cells, 32,768 gain bits written per load in 0.253 ns (31.6 ps per row, P), refreshed every 128 passes (0.76 μs, D).

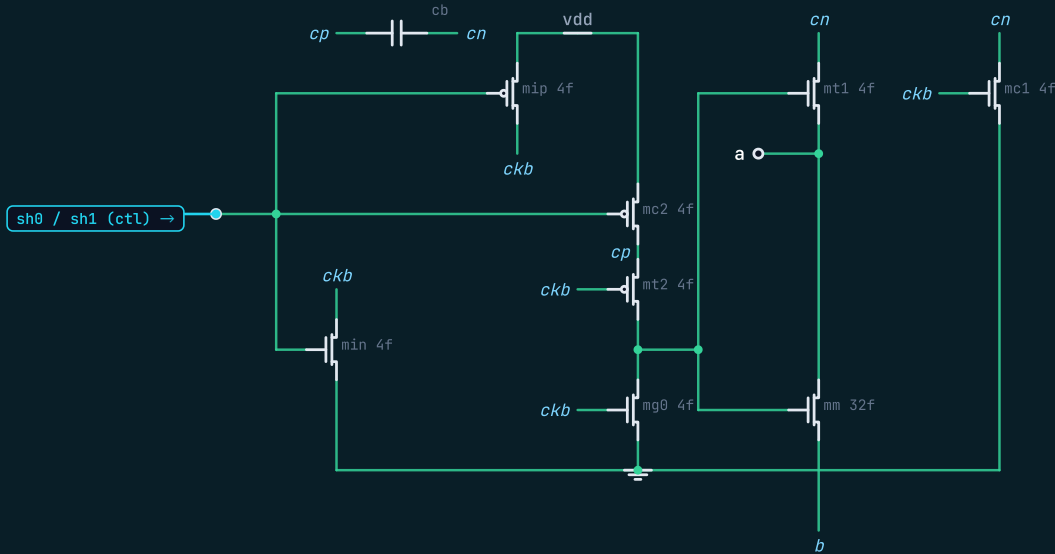
block views drawn at 2 rows × 3 columns; supplies left off the blocks

● B3 column: share switch, ping-pong bank, phase decode

One bank of one column side: bootstrapped share, 1:16 merge, the step caps that are the SAR's C-DAC, and the phases that run it.

bsw · bootstrapped share switch

constant V_GS: mm's gate rides cb on top of the input

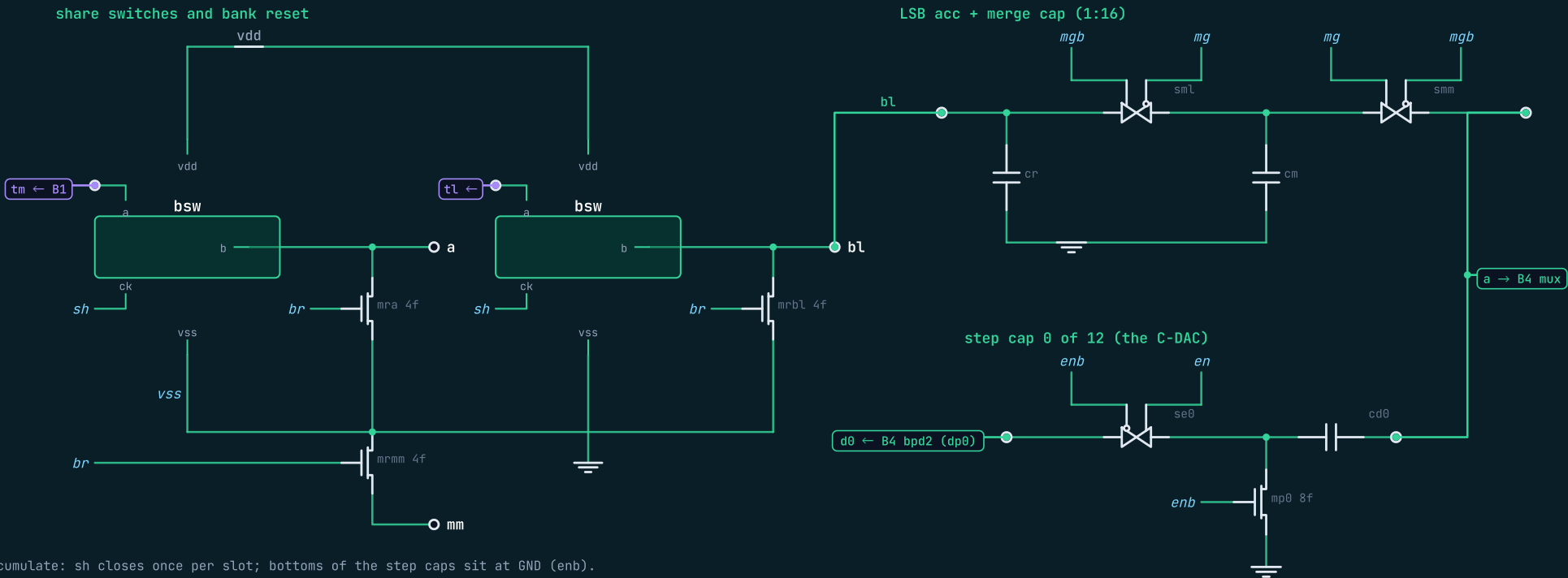


a ← tm / tl (the slice top plate) · b → a / bl (the bank's accumulator).
ck low: c1, c2 charge cb to VDD, g0 holds the gate at 0.
ck high: t1 / t2 put cb between the input and mm's gate: V_GS ≈ 0.65 V.
mm 32 fins; cb 20 fF » its gate (V7: the gate reaches about 1.3 V).

netlist: bsw · 2 per bank (MSB, LSB)

bank · one accumulation bank of one column side

share → acc · 1:16 slice merge · the MSB acc is the C-DAC of the direct SAR

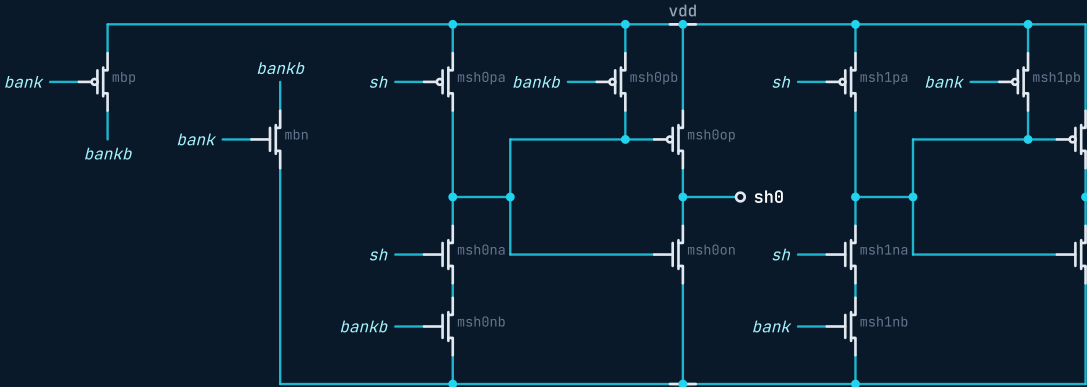


Accumulate: sh closes once per slot; bottoms of the step caps sit at GND (enb).
Merge, under the next pass: mg joins C_m (p = 0.134, 1:16) to the MSB acc a.
Convert: en hands each step cap's bottom to the converter's driver d<k>.
Reset: br clears a, bl and mm before the bank's next pass (ψ_brst).
Step caps: 1024, 512, 256, 128, 64, 32, 32, 16, 8, 4, 2, 1 LSB of the 56 fF slice.

netlist: bank · × 2 ping-pong × 2 sides per column

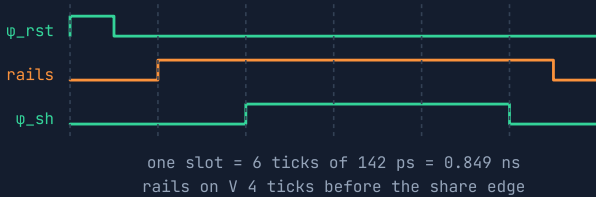
ctl · per-bank phase decode

from imc_seq's sh, mrg, brst and bank



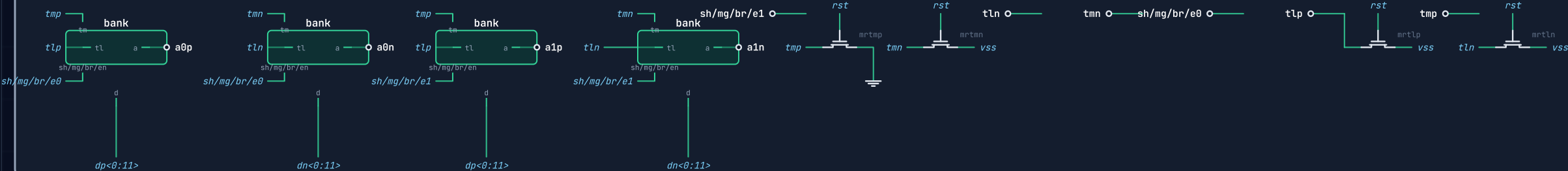
netlist: ctl · drawn: sh0, sh1 (mg, br alike)

BS6H slot and the bank hand-off (imc_seq, RTL)



pass p: 7 slots on bank p mod 2 (42 ticks, 5.96 ns)
hand-off: bank flips one tick after the last share;
ψ_mrg merges the old bank under the next pass's slot 0,
ψ_samp hands it to the SAR; ψ_brst resets the new bank
in slot 0, tick 1. The FSM waits only if the converter
has not sampled the previous bank yet (StWait).

col · one column pair (cktImg block view): 4 top-plate resets + 4 banks



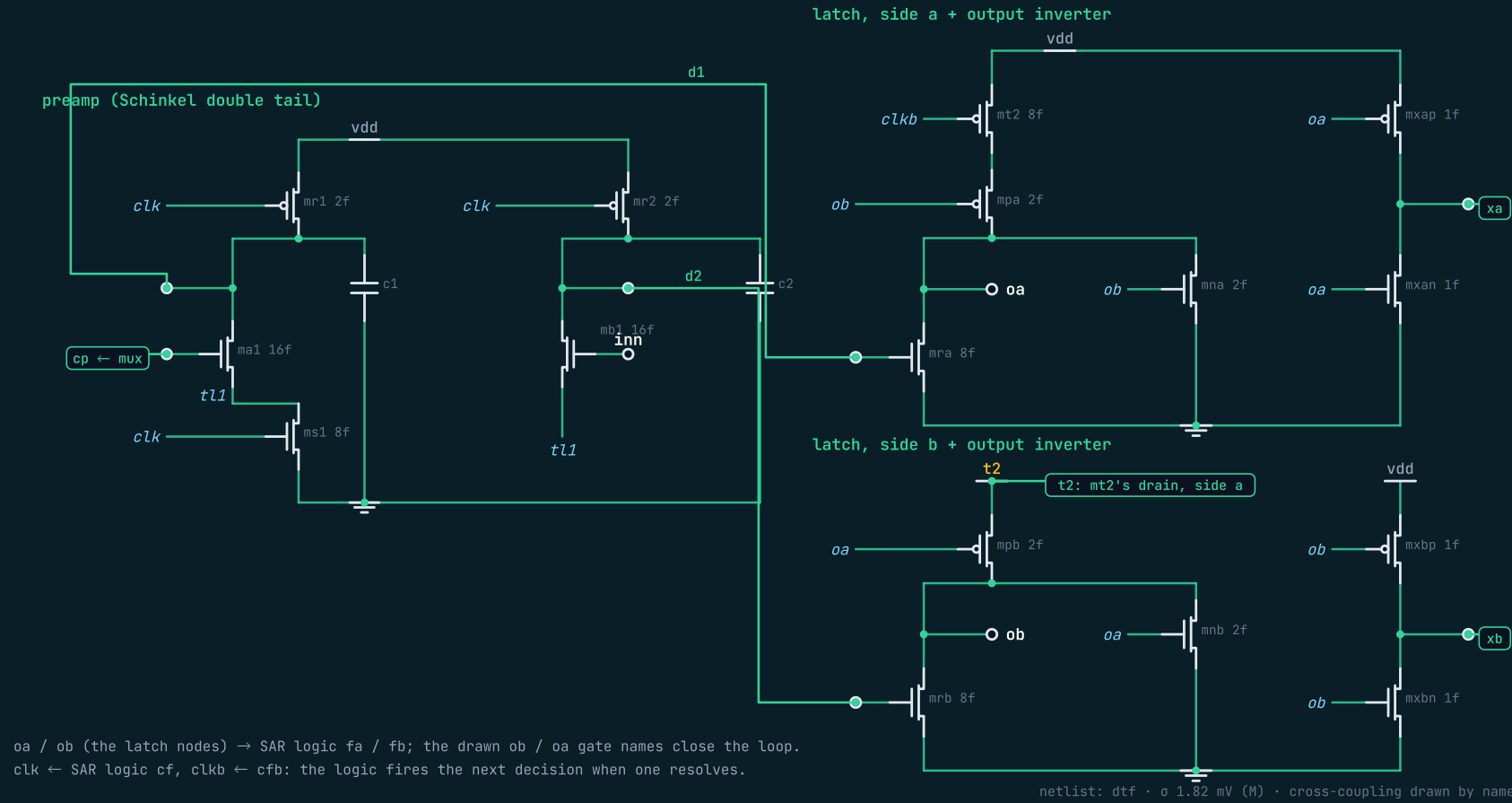
netlist: col · × 256 per tile

● B4 E-trim SAR converter

Mux → fast and quiet double-tail comparators (drawn by stage) → asynchronous SAR logic → 3-level drivers on the bank's step caps.

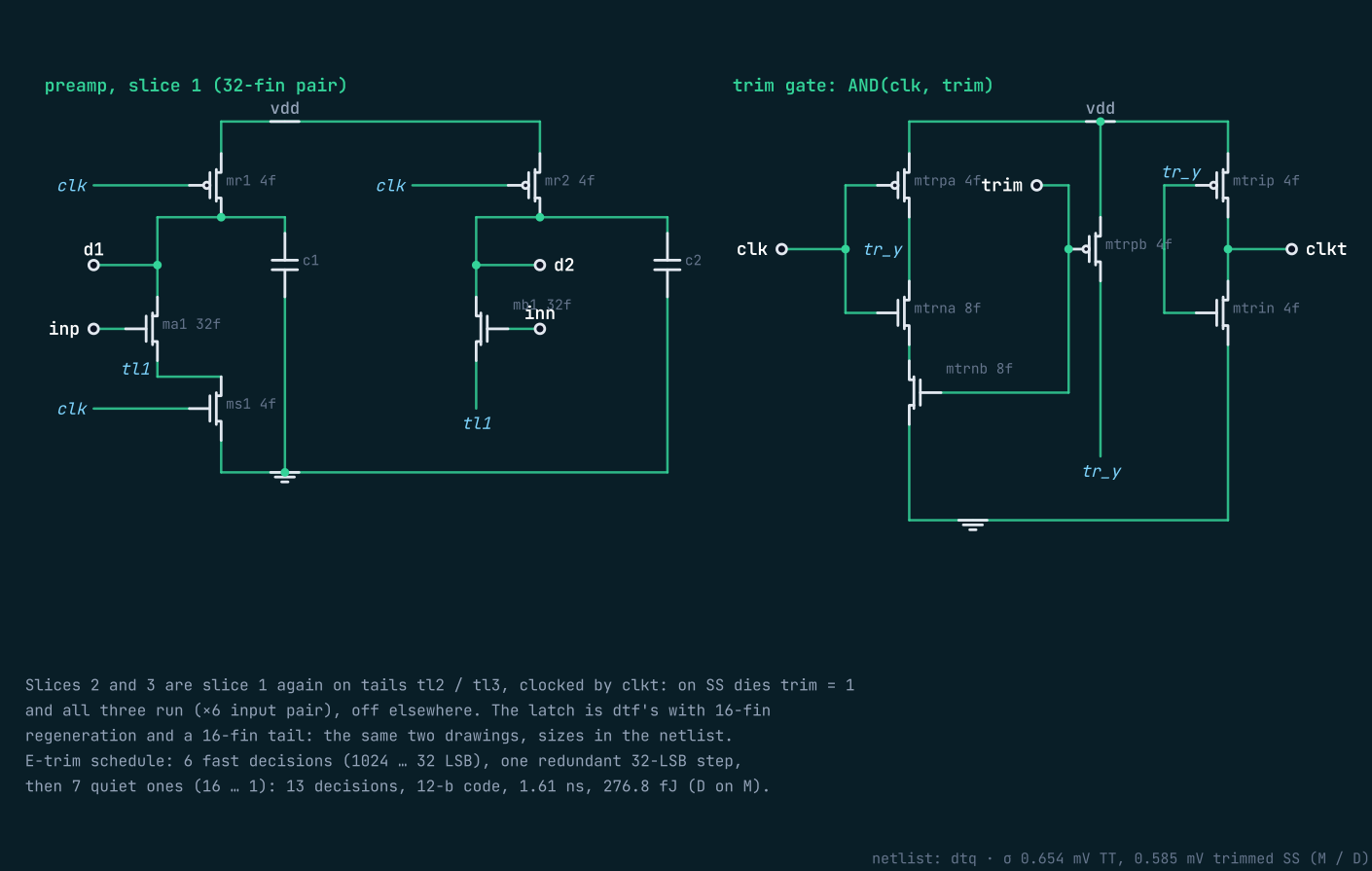
dtf · fast double-tail comparator (decisions 1-6)

stage 1: tail + input pair on Di nodes d1 / d2, reset by clk · stage 2: latch tailed by mt2 on clk_b

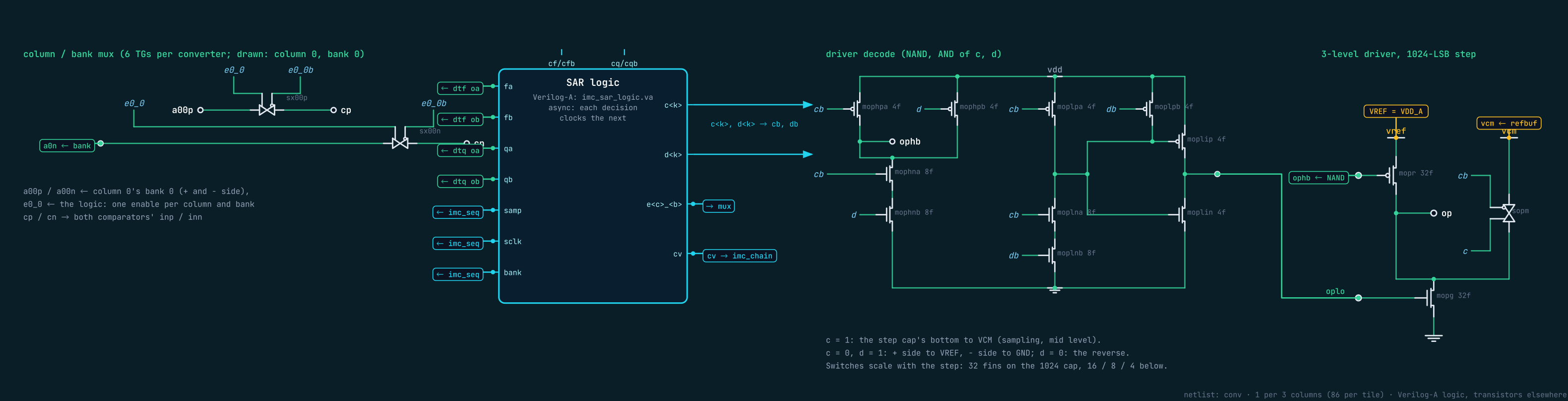


dtq · quiet tail-starved double-tail (decisions 8-13)

three ×2 input slices on the shared d1 / d2; slices 2-3 run on clk_t = clk · trim



conv · one converter: bank mux, comparators, SAR logic, 3-level C-DAC drivers

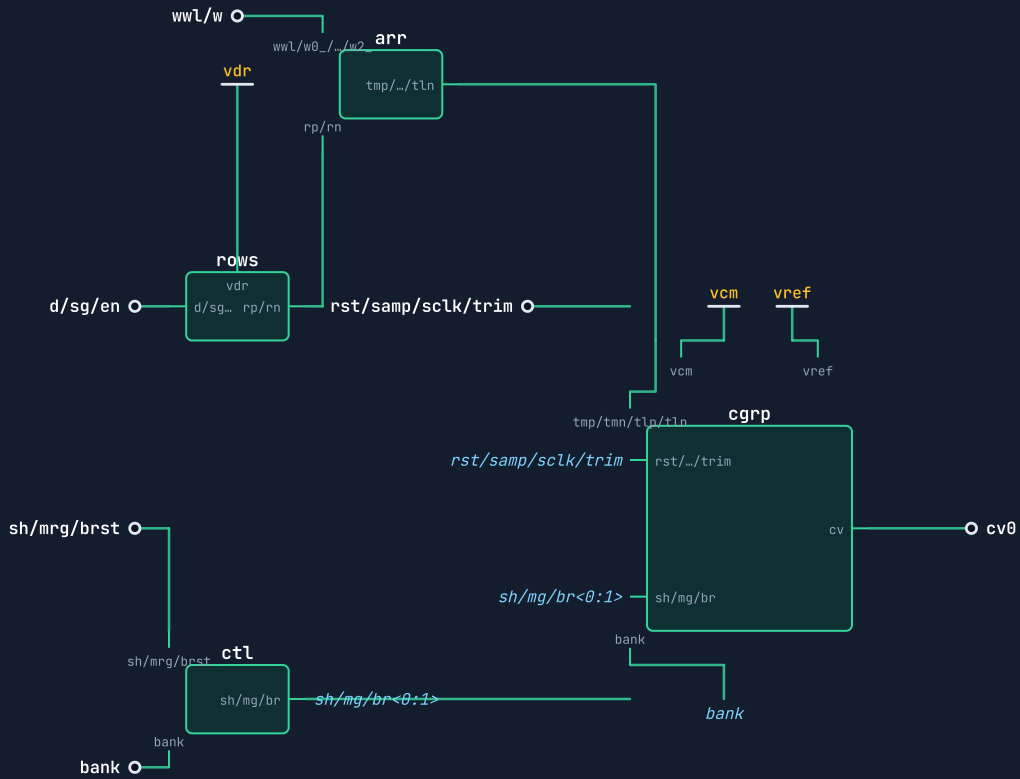


● Netlist hierarchy and the B5 reference

cktImg block views of the tile and the tile pair (buses bundled, supplies off the blocks), the ADC-share-3 converter group, and the class-A VCM follower.

tile (cktImg block view)

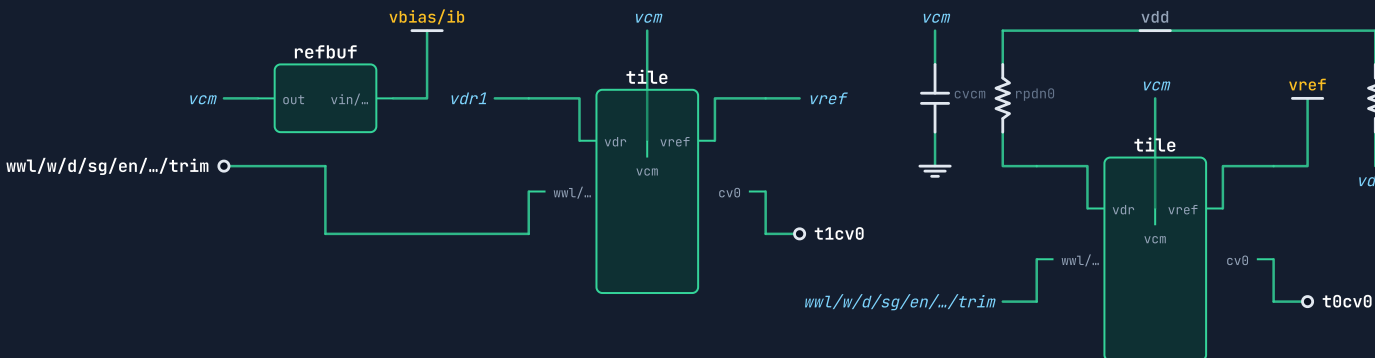
rows → arr → cgrp, with ctl decoding the bank phases



netlist: tile · 2 of 8 rows x 3 of 256 columns, 1 converter

tiles (cktImg block view)

N tiles on their R_PDN, one VCM buffer and decap

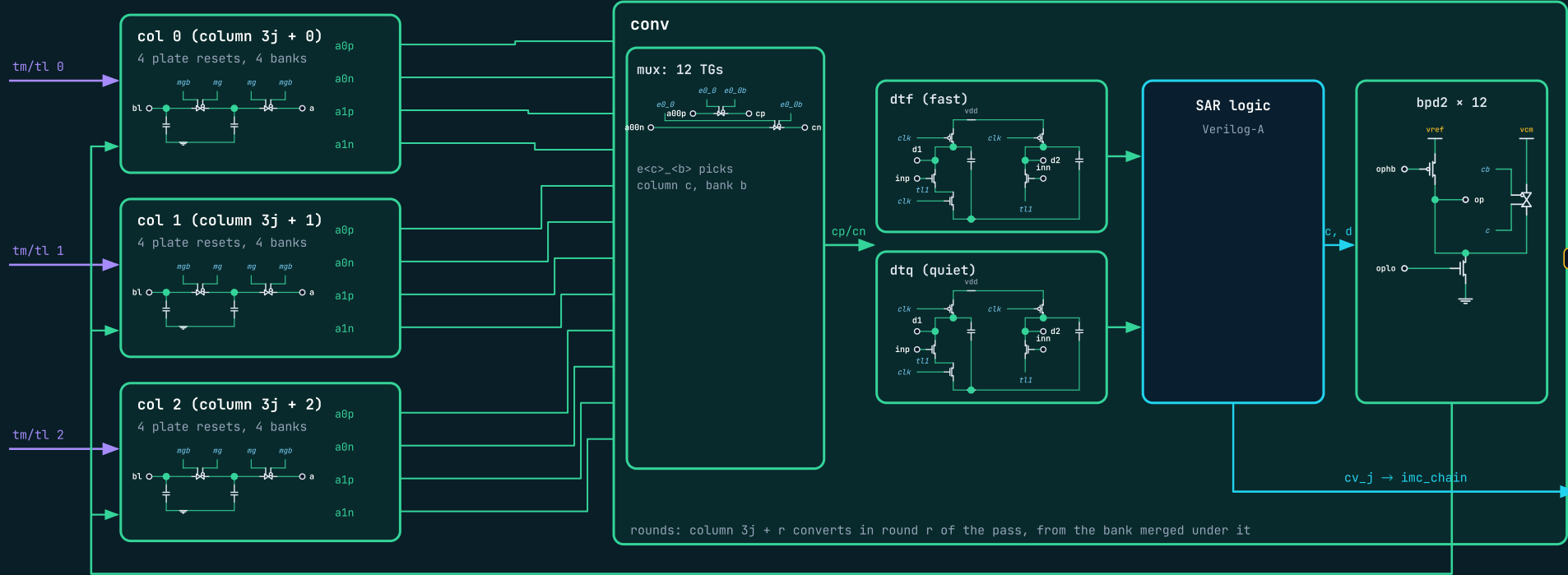


rpdn<t>: the tile supply's PDN resistance (0.4 Ω per full tile) between vdd and vdr<t>.
cvcm: 20 pF of VCM decap per tile. refbuf: the class-A follower that holds vcm.
Every tile sees the same phases; the RTL gives each its own row codes and weights.

netlist: tiles · 2 of the die's tiles

cgrp · ADC share 3: three columns, one converter

12 bank nodes (3 columns × 2 banks × 2 sides) into one converter; tm/tl c: column c's 4 slice top plates

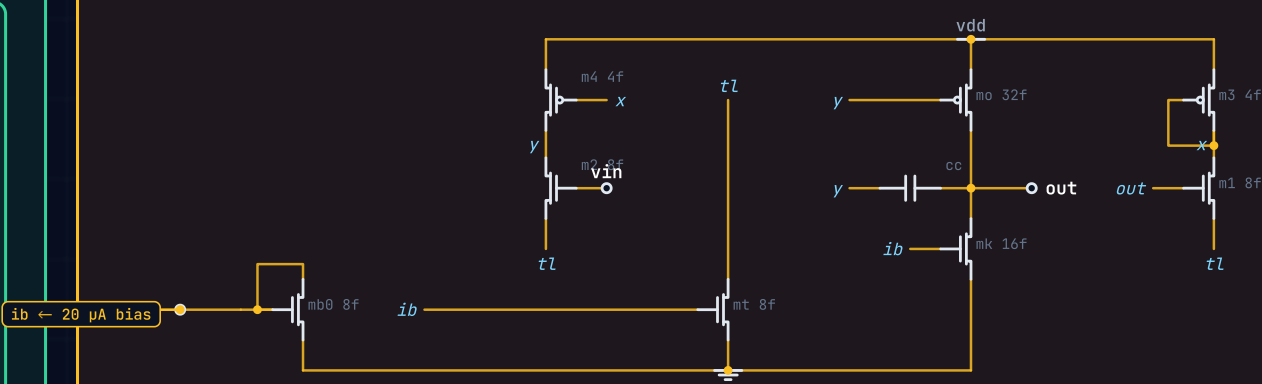


dp<0:11> / dn<0:11>: the drivers onto every bank's step caps (the C-DAC)

netlist: cgrp = 3 × col + conv · × 86 per tile

B5 · refbuf: class-A VCM follower

two-stage Miller, unity gain, the loop drawn by name (out on m1's gate)



vin ← vbias (VDD_A / 2); out → vcm, the converters' mid level.
m1 / m2 input pair on tail mt, m3 / m4 mirror, mo + mk class-A output, cc 60 fF Miller. The feedback goes to the mirror's diode side because the PMOS output stage inverts (found in the transistor run, M).
Reference droop: 130 μV per conversion against 3.7 μV (FAIL by 35×),
a calibrated static error: the ref term is 85 dB (D).

netlist: refbuf

● Digital: imc_driver RTL and the rail

The tile-array controller's modules (digital/imc_driver/src) and the attention rail the spec adds.

imc_driver: the controller a weight-stationary systolic array would have, mapped onto the charge-domain tile · Verilog-2001 (it also runs as a VerA device in ESPice)

imc_driver.v

top: descriptor generator, row codes

descriptor schedule

for n-block, m-chunk, k-group, token: one element;
element e enters tile t at pass e + t + 1 (systolic skew)

activation read

one word per tile per pass: 8 × INT8 + block-8
scale byte, address token·6·N + k·N + t

row codes (BS6H)

|x| ≤ 127 (−128 → 127), sign picks rp / rn;
slot s drives bit s; a flop per pin, cut by drv_cut

hand-off / sample capture

the element of each bank at hand-off and at
the sample, for the chain's alignment

imc_seq.v

pass FSM and phase generator

pass FSM

Idle → Drive (7 slots × 6 ticks) → Write (on weight change
or refresh) / Stall (weights late) / Wait / Done

merge engine

takes the driven bank at hand-off; φ_mrg 2 ticks under
the next pass's slot 0, then φ_samp when the SAR is free

conversion counter

AdcShare rounds of 13 ticks on its own counter;
cap / cap_round / conv_done to the chain

phase flops

φ_rst, φ_sh, φ_mrg, φ_brst, φ_samp, sar_clk, bank:
every pin a single flop, edges on the DLL tick

imc_wstage.v

weight staging and the array writer (B1 write port)

stage banks

double-buffered per tile: the HBM stream fills bank
g mod 2 one group ahead of the array

array writer

WL one row per tick in the write window, WBL per
column: [7] sign, [6:0] |w| (sign-magnitude cell)

refresh

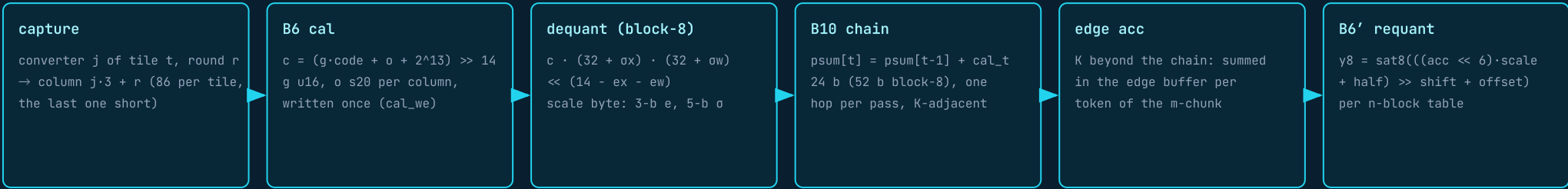
a group in the array for RefreshPasses (128 = 0.76 μs)
is rewritten from its bank (retention 1.32 μs, P)

ready / stall

ready when every needed bank is staged; else the
sequencer stalls and counts the ticks

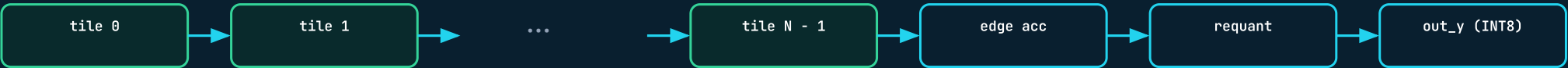
imc_chain.v

code capture, B6 calibration, block-8 dequant, B10 accumulator chain, B6' requant



Alignment: tile t converts element e at pass e + t + 1, so at each conversion-done the value
tile t - 1 left in its psum register is the element tile t just converted (bit-exact, 25 cases).
Ports: codes_i (N·86·12 b), desc_cv_i / xs_i (the element and its x scale), ws_data_i (w scales),
cal_*, rq_data_i → out_valid_o, out_y_o (INT8), out_acc_o.

Checks (M on the RTL): t_pass 5.964 ns (42 ticks); no stall after fill; GEMV at the per-tile
HBM share: 28,144 ticks against the 28,032 law; driver + ideal tiles bit-exact against the golden.



psum[t] = psum[t - 1] + cal_t(code): one hop per pass, exact; only completed outputs leave the chain

Digital rail (spec ARCH_CHOSEN B8)

not in RTL yet: the blocks the spec names

attention lanes

Q·K^T and P·V: q8 × k4 bulk,
q8 × k8 for sink 8 + recent 120

online softmax

base-2 exp from a ROM

FWHT

Hadamard on q, k, v, o (the 4.5 dB credit)

RMSNorm · RoPE · SwiGLU

per token, INT8 in and out

KV pack / unpack

4-b codes + fp16 scale per (token, head);
4-b ring residual, dropped on exit